2019-DSE
BIO

PAPER 1A

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY

HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2019

BIOLOGY PAPER 1

8:30 am – 11:00 am (2 hours 30 minutes)
This paper must be answered in English

GENERAL INSTRUCTIONS

(1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.

(2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book **B**.

(3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book **B**. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

(1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.

(2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.

(3) All questions carry equal marks.

(4) **ANSWER ALL QUESTIONS**. You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.

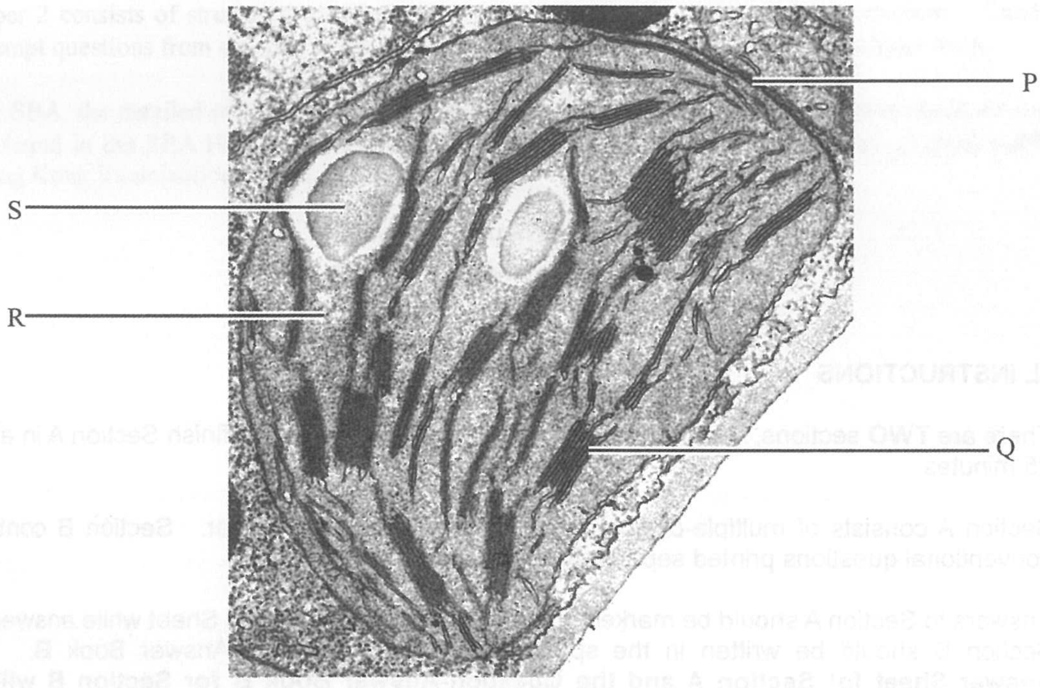
(5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.

(6) No marks will be deducted for wrong answers.

Not to be taken away before the end of the examination session

There are 36 questions in this section.
The diagrams in this section are NOT necessarily drawn to scale.

Directions: Questions 1 and 2 refer to the electron micrograph below, which shows a chloroplast of a plant cell:



1. During photosynthesis, light is captured at

   A. P.
   B. Q.
   C. R.
   D. S.

2. Carbon dioxide with radioactively labelled oxygen was provided to the plant cell for photosynthesis. Radioactivity can be detected in

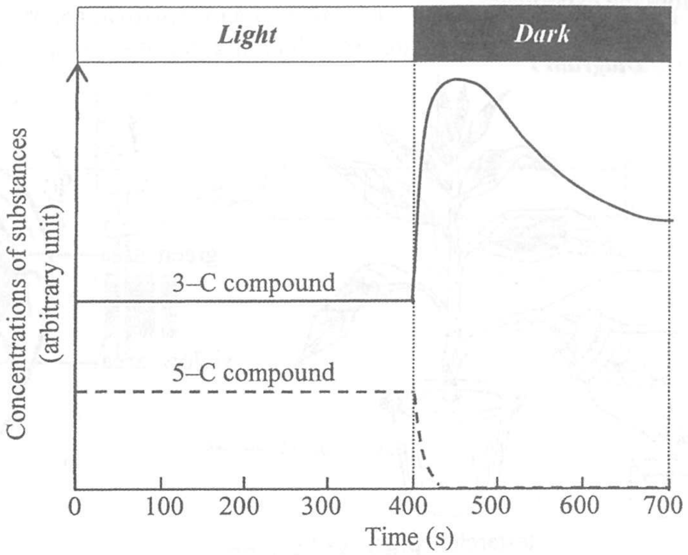
   A. oxygen produced at Q.
   B. oxygen produced at R.
   C. glucose produced at Q.
   D. glucose produced at R.

3. Which of the following process(es) in the respiratory pathways release(s) carbon dioxide?

   (1) Oxidative phosphorylation
   (2) Reactions in the Krebs cycle
   (3) Conversion of glucose to pyruvate

   A. (1) only
   B. (2) only
   C. (1) and (3) only
   D. (2) and (3) only

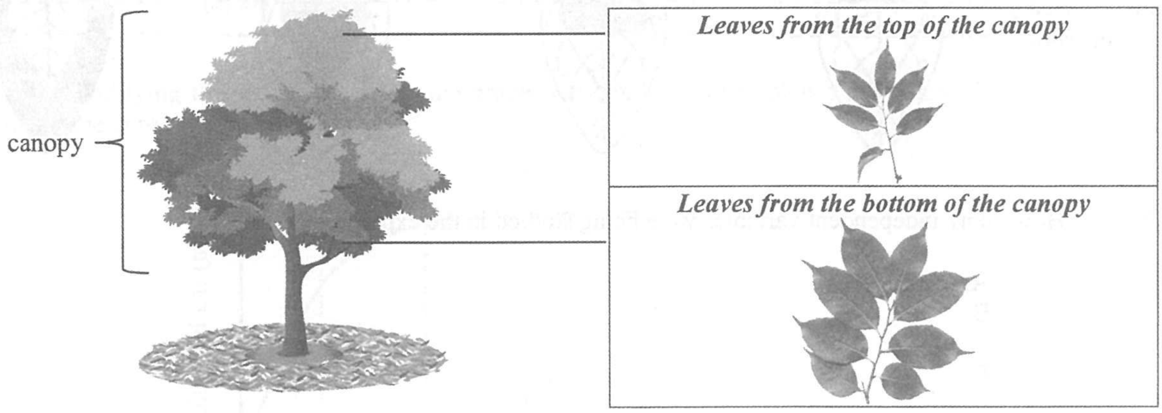
4. The graph below shows the changes in the relative concentrations of a 3–C compound and a 5–C compound (carbon dioxide acceptor) in the Calvin cycle in green plant cells which have been kept in bright light and then in darkness.



Which of the following is ***not*** a reason why the concentration of the 5–C compound decreased in the dark?

A. The 5–C compound was converted to the 3–C compound.
B. The 5–C compound combined with carbon dioxide to form glucose directly.
C. Regeneration of 5–C compound stopped because there was no ATP from photochemical reactions.
D. Regeneration of 5–C compound stopped because there was no NADPH from photochemical reactions.

5. The photographs on the right below show leaves taken from different parts of the canopy of the same tree. (Note: The photographs are of the same magnification.)



Which of the following is the most likely explanation for the differences between the leaves taken from the two parts of the canopy?

A. The leaves from the top of the canopy are smaller because they do not receive sufficient water for growth.
B. The leaves from the top of the canopy are smaller because they can reduce water loss due to transpiration.
C. The leaves from the bottom of the canopy are larger because they can store more food from photosynthesis.
D. The leaves from the bottom of the canopy are larger because they can collect light escaped through the top of the canopy.

Directions: Questions 6 and 7 refer to Diagram I and Diagram II below. Diagram I shows a set-up prepared by a student to study the conditions for photosynthesis. Diagram II shows the leaf surface of a variegated leaf X before the experiment.

Diagram I

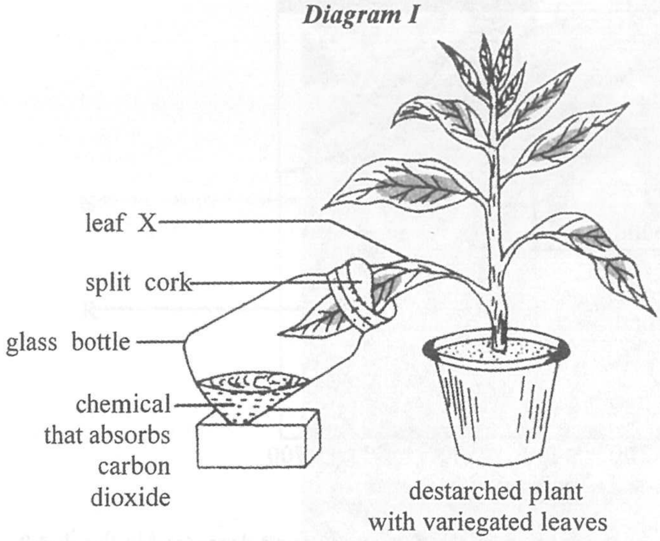
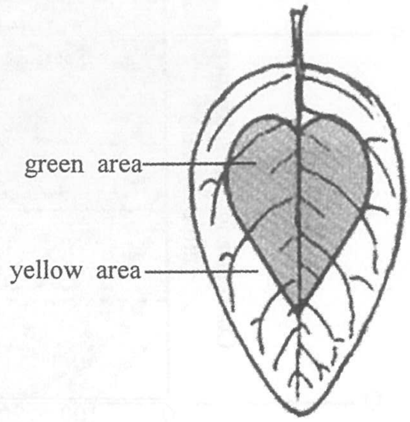


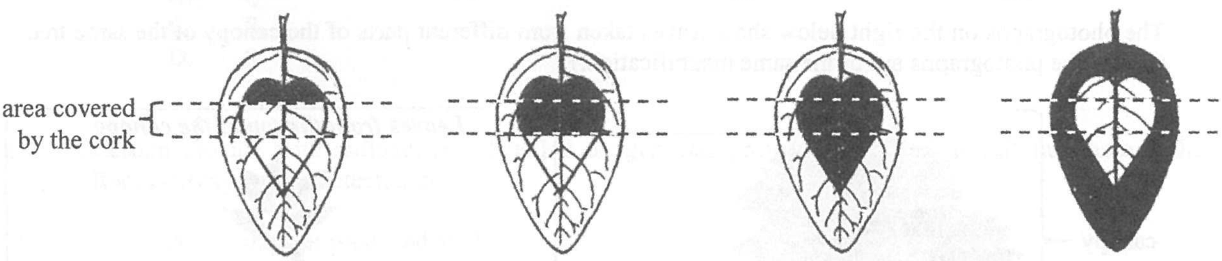
Diagram II



6. After leaving the set-up under sunlight for several hours, iodine test was carried out on leaf X. Which of the following diagrams correctly shows the results?

Key: brown / blue black

A. B. C. D.



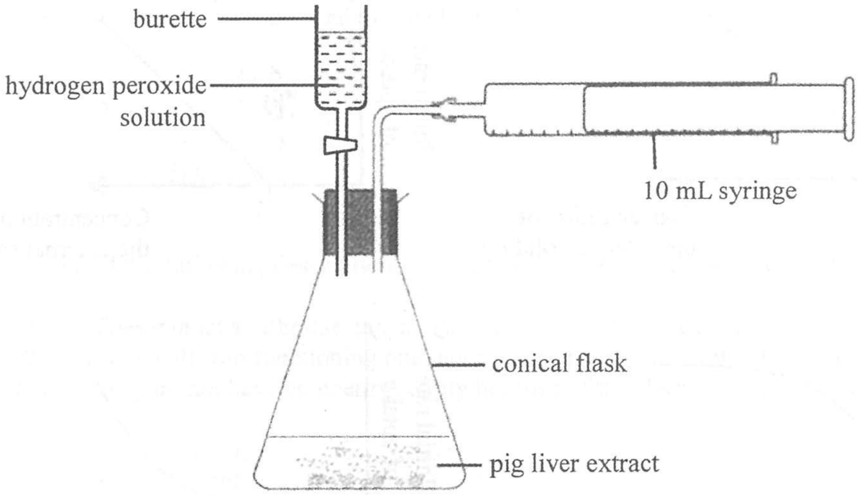
7. How many independent variables were being studied in the experiment?

A. 1
B. 2
C. 3
D. 4

8. According to the requirements of various methods of transport across the cell membrane, which of the following combinations is correct?

	Requirements		
	Energy input	***Membrane protein***	***Concentration gradient***
A.	phagocytosis	active transport	osmosis
B.	diffusion	osmosis	active transport
C.	active transport	phagocytosis	phagocytosis
D.	osmosis	diffusion	diffusion

Directions: Questions 9 and 10 refer to the diagram below, which shows an experimental set-up prepared by a student to investigate the effect of temperature on catalase activity. Pig liver extract contains catalase which speeds up the breakdown of hydrogen peroxide into oxygen and water. A fixed volume of hydrogen peroxide solution was added to the liver extract and a 10 mL syringe was used to collect the oxygen gas released from the reaction mixture.

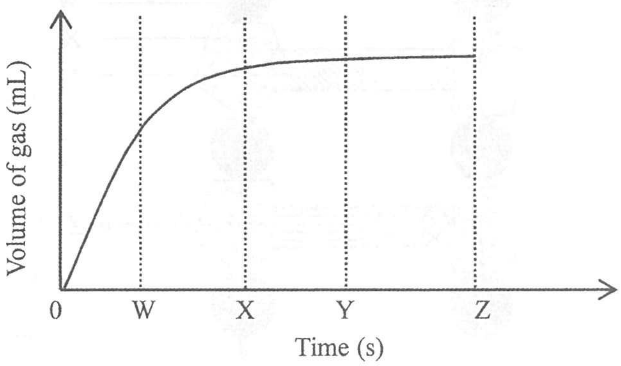


9. In the trial run conducted at room temperature, the student found that the volume of oxygen released was greater than the maximum collection volume of the syringe. How should he modify the set-up in order to collect valid data when repeating the experiment at different temperatures?

(1) use a larger syringe
(2) use a larger conical flask
(3) reduce the volume of the hydrogen peroxide solution added

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

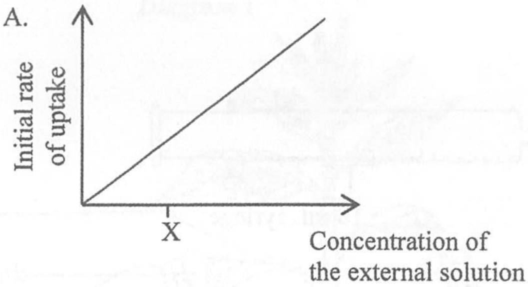
10. After modifying the set-up, the following graph was obtained which shows the volume of gas collected over time at room temperature:

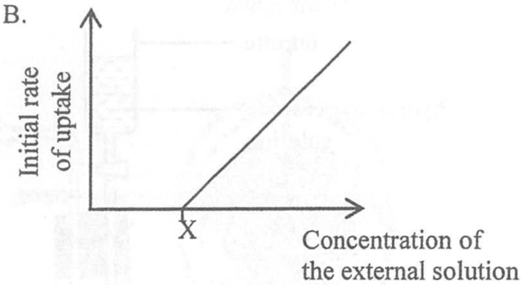


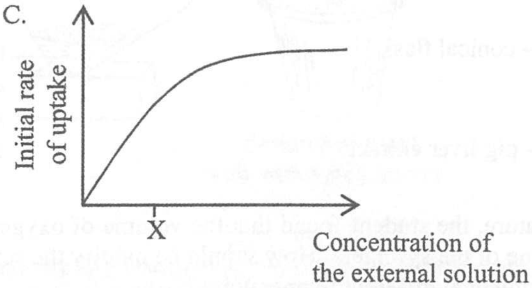
The student planned to use the volume of gas collected over a fixed period of time as the dependent variable to study the effect of different temperatures on catalase activity. Which of the following is the most suitable time period for the measurement?

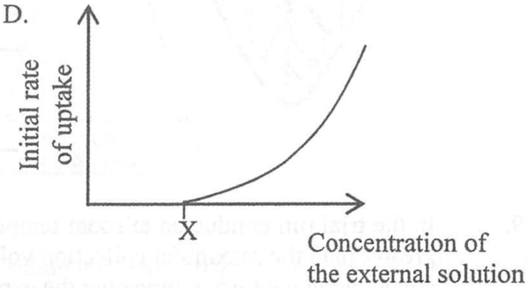
A. 0 – W
B. 0 – X
C. 0 – Y
D. 0 – Z

11. Solutions of different concentrations of a solute with a small molecular size were prepared and some plant cells were immersed in each of the solutions. Which of the following graphs shows the initial rate of uptake of the solute by means of diffusion?
(Note: X is the concentration of the solute inside the plant cells)

A.


B.


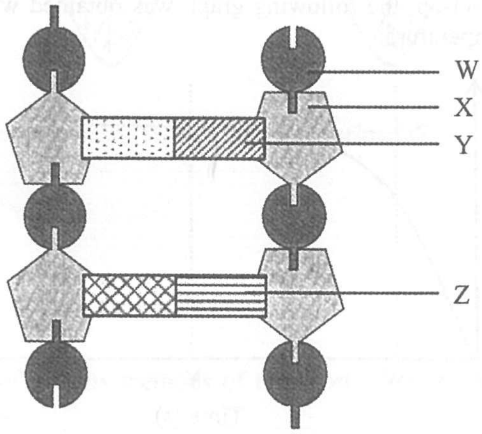
C.


D.


12. Which of the following statements about the human egg and sperm is correct?

A. Both have the same number of genes.
B. Both have the same amount of cytoplasm.
C. Both have the same amount of food reserve.
D. Both have the same number of chromosomes.

13. The diagram below shows a DNA model:



Which of the following combinations shows the most probable identities of molecules W, X, Y, and Z?

	W	*X*	*Y*	*Z*
A.	sugar	phosphate	cytosine	thymine
B.	sugar	phosphate	cytosine	guanine
C.	phosphate	sugar	adenine	thymine
D.	phosphate	sugar	adenine	guanine

14. In a family, the father is red-green colour blind (a recessive X–linked trait) and is of blood group A, while the mother has normal vision and is of blood group B. Which of the following could be the phenotypes of their biological child?

(1) a normal-vision girl with blood group O
(2) a red-green colour blind girl with blood group O
(3) a red-green colour blind boy with blood group AB

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

15. Which of the following description(s) of human red blood cells is/are correct?

(1) They cannot synthesise enzymes or proteins for repair because of the absence of the nucleus.
(2) They will stop functioning one day because haemoglobin will be used up.
(3) They do not have an energy supply because of the absence of mitochondria.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

16. Peter wanted to throw a crumpled paper into the rubbish bin in a dim room. He found that he could see the bin more clearly if he tried to focus on objects right next to the bin. Which of the following statements about the image formation of the bin is correct?

A. Image of the bin is formed on the yellow spot where there are cone cells only.
B. Image of the bin is formed on the yellow spot where there are more cone cells than rod cells.
C. Image of the bin is formed on the periphery of the retina where there are rod cells only.
D. Image of the bin is formed on the periphery of the retina where there are more rod cells than cone cells.

17. Wearing contact lenses for too long will have an adverse effect on the eyes because this decreases the amount of oxygen reaching the eyes. Which of the following eye structures is most affected in this case?

A. iris
B. lens
C. cornea
D. sclera

18. Which of the following organs serves both endocrine and exocrine functions?

A. pancreas
B. pituitary
C. oesophagus
D. adrenal gland

Directions: Questions 19 and 20 refer to Diagram I and Diagram II below. Diagram I shows a leg and its associated muscles while Diagram II shows a woman practising yoga.

Diagram I

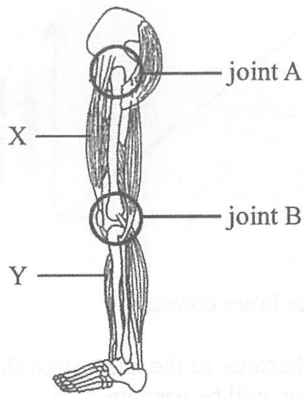
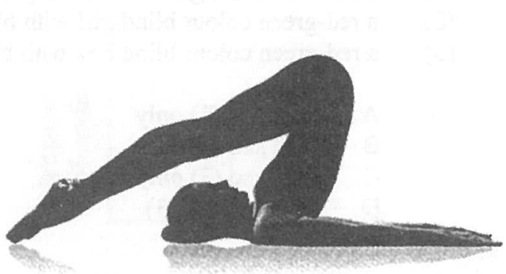


Diagram II



19. Which of the following combinations correctly identifies joints A and B in Diagram I?

	Joint A	*Joint B*
A.	Hinge joint	Hinge joint
B.	Hinge joint	Ball and socket joint
C.	Ball and socket joint	Ball and socket joint
D.	Ball and socket joint	Hinge joint

20. Which of the following combinations correctly indicates the state of muscles X and Y when the woman is maintaining the yoga posture shown in Diagram II?

	X	*Y*
A.	contracted	contracted
B.	contracted	relaxed
C.	relaxed	contracted
D.	relaxed	relaxed

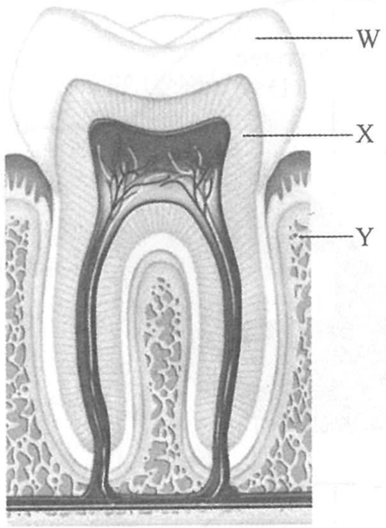
21. Daisy is very hungry. When the waiter puts her favourite dish on the table, her saliva secretion increases. Which of the following body parts controls this response?

A. cerebrum
B. cerebellum
C. salivary gland
D. medulla oblongata

22. When ventricles contract, the valves between ventricles and atria close. Which of the following is the cause of the valve closure?

A. The heart tendons hold the valves in position.
B. The refilling of blood at the atria pushes the valves so that they close.
C. The closure of valves prevents the blood from flowing back to the atria.
D. The higher blood pressure resulting from ventricular contraction pushes the valves so that they close.

Directions: Questions 23 and 24 refer to the diagram below, which shows a section of one type of tooth and its associated structures:



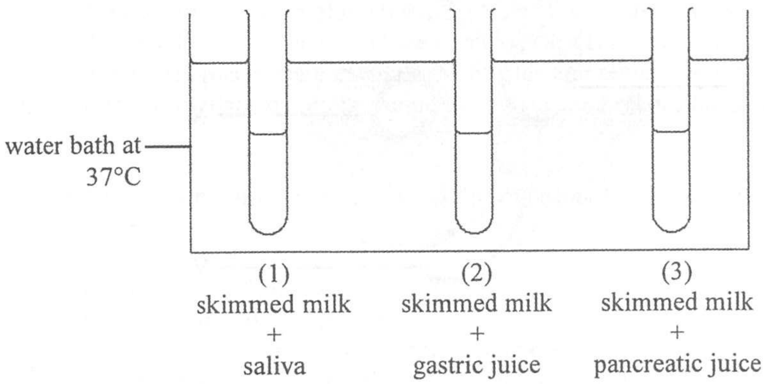
23. Which of the following are living tissues that contain a large amount of calcium salt?

A. W and X
B. W and Y
C. X and Y
D. W, X and Y

24. The number of this type of tooth in the milk dentition is

A. 0.
B. 4.
C. 8.
D. 12.

25. The diagram below shows a water bath with three test-tubes containing different mixtures:

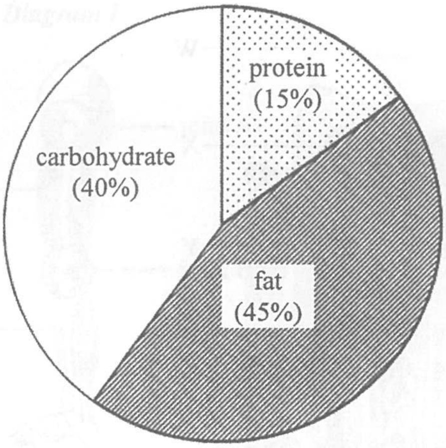


Chemical digestion of food takes place in test-tubes

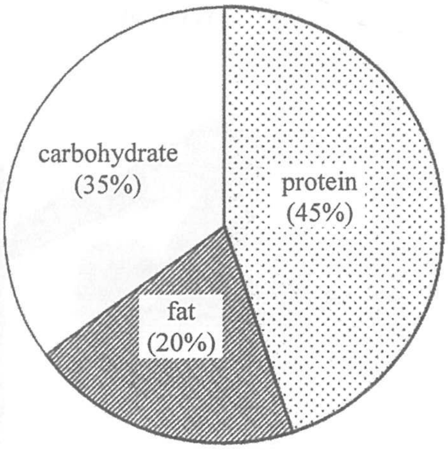
A. (1) and (2) only.
B. (1) and (3) only.
C. (2) and (3) only.
D. (1), (2) and (3).

26. The following charts show the composition of four different foodstuffs. Which foodstuff yields the highest amount of energy per gram?

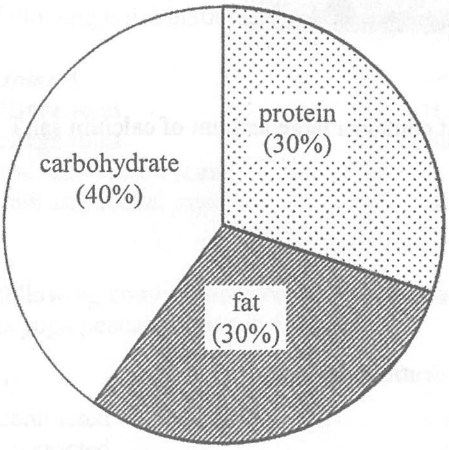
A.



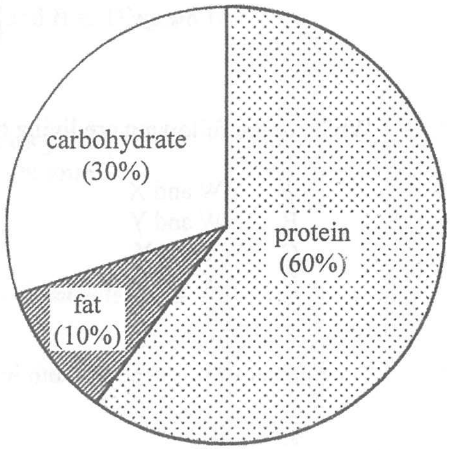
B.



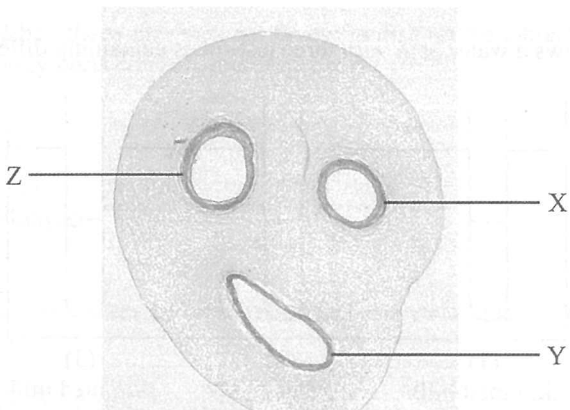
C.



D.



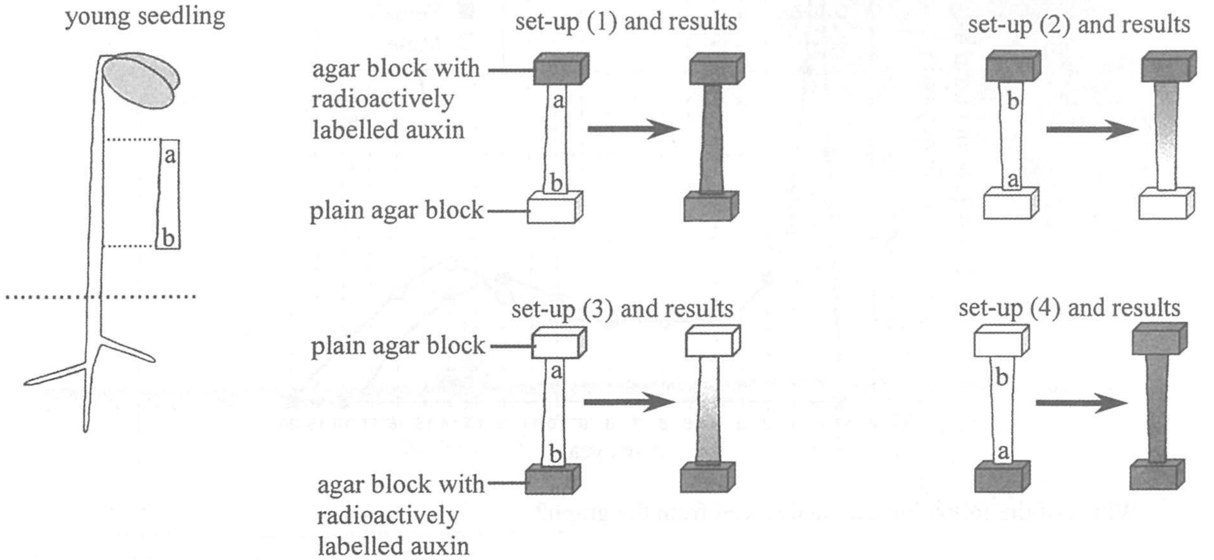
27. The photomicrograph below shows a section of a human umbilical cord with three blood vessels:



Which of the following comparisons of the content of the blood vessels is correct?

A. The blood in vessel X has a higher oxygen content than that in vessel Y.
B. The blood in vessel Z has a higher glucose content than that in vessel X.
C. The blood in vessel Y has a higher amino acid content than that in vessel X.
D. The blood in vessel Y has a higher carbon dioxide content than that in vessel Z.

28. To investigate the transport of auxins in the stem of a young seedling, a plain agar block and an agar block soaked in radioactively labelled auxins were prepared and placed at different ends of a cut stem, as shown in the diagram below. The relative amounts of radioactivity in different parts are shown as different intensities of shaded area.



Which of the following conclusions can be drawn from the above results?

(1) Transport of auxins in the stem is not affected by gravity.
(2) Transport of auxins in the stem involves an active process.
(3) Transport of auxins in the stem mainly takes place from a to b.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

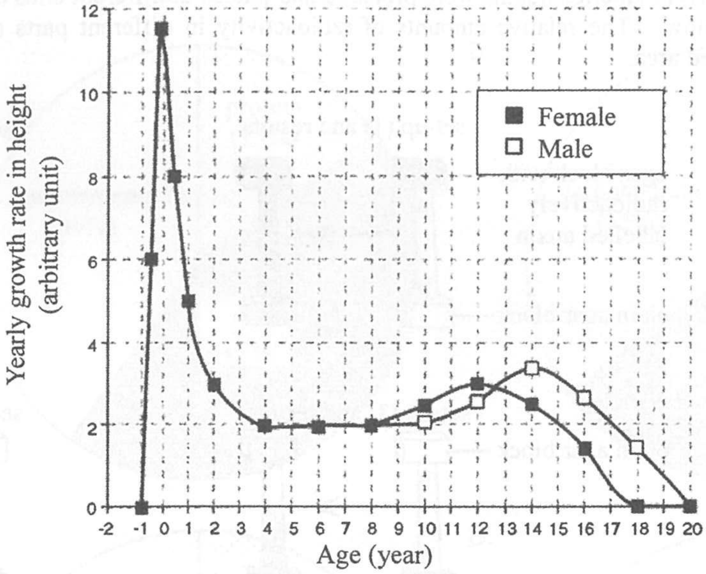
29. If a student wants to find out whether the distribution of plant species is affected by the slope of a hillside, which of the following sampling methods should be used?

A. Set up a line transect along the hillside and record the plant species that touch the line.
B. Set up a line transect around the hillside and record the plant species that touch the line.
C. Randomly place quadrats along the hillside and record the plant species within the quadrat.
D. Randomly place quadrats around the hillside and record the plant species within the quadrat.

30. Which of the following parameters is best used for recording the growth of a potted germinating seedling over a period of time?

A. the dry mass of the seedling
B. the fresh mass of the seedling
C. the total surface area of the seedling's leaves
D. the time taken for the seedling's first leaf to appear

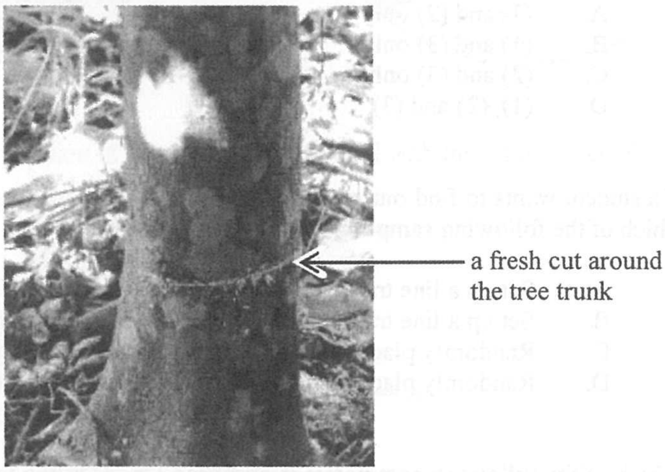
31. The graph below shows the growth rate of humans:



Which of the following can be deduced from the graph?

A. The brain grows rapidly from age 0 to age 4.
B. There is no more change in growth for males after age 20.
C. The duration of adolescence is the same in males and females
D. Growth of the reproductive organs begins at age 10 in females.

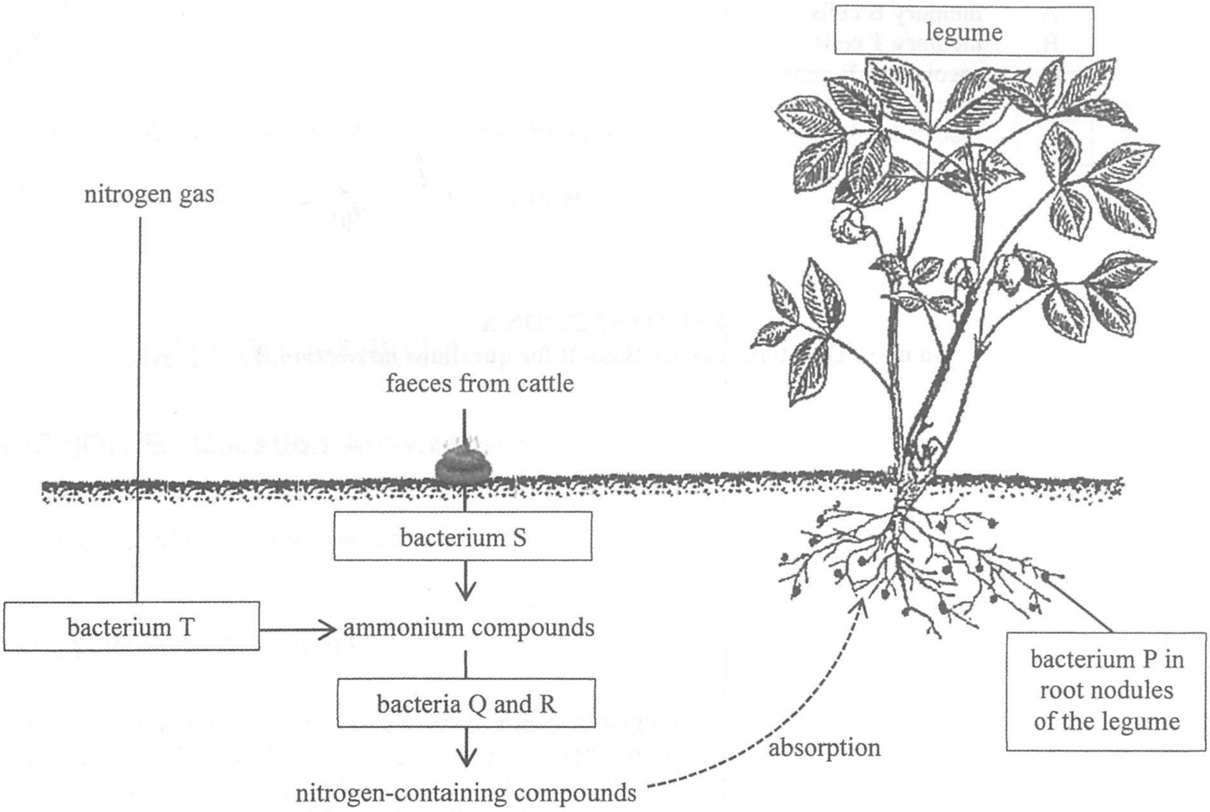
32. The photograph below shows a fresh cut around the trunk near the bottom of a tree:



The tree eventually died. Which of the following is the most likely reason for the death of the tree?

A. Water could not be transported to the leaves for transpiration.
B. Water could not be transported to the leaves for photosynthesis.
C. Minerals could not be transported upward for protein synthesis.
D. Photosynthetic products could not be transported to the roots for respiration.

Directions: Questions 33 and 34 refer to the diagram below, which shows some processes of the nitrogen cycle:



33. Which of the following statements about the bacteria shown in the diagram is correct?

    A. P belongs to parasitic bacteria.
    B. Q belongs to nitrifying bacteria.
    C. R belongs to nitrogen-fixing bacteria.
    D. S belongs to denitrifying bacteria.

34. Which of the following bacteria has a similar role to fungi in the cycling of materials?

    A. Q
    B. R
    C. S
    D. T

35. Type II diabetic patients may feel dizzy after prolonged exercise. Which of the following is the most likely explanation for this?

    A. Their blood glucose level has dropped to too low a level because they do not have enough stored glycogen to replenish the glucose used.
    B. Their blood glucose level has dropped to too low a level because they do not have enough glucagon to stimulate the conversion of glycogen to glucose.
    C. Their blood glucose level has risen to too high a level because they keep losing water during exercise.
    D. Their blood glucose level has risen to too high a level because they do not have enough insulin to stimulate the conversion of glucose to glycogen.

36. Antibodies are produced by

A. memory B cells.
B. memory T cells.
C. specialised B cells.
D. specialised T cells.

END OF SECTION A
Go on to Question-Answer Book B for questions on Section B